

MUHAMMAD MAHEEM

AI Engineer & Full-Stack Machine Learning Developer

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SUMMARY

AI engineer focused on production LLM systems — RAG pipelines, evaluation frameworks, and full-stack AI products built with FastAPI and Next.js. Shipped 15+ projects spanning legal-tech contract analysis, HR resume screening, document intelligence, and computer vision, owning each system end-to-end from retrieval/model design through deployment and observability.

EDUCATION

Superior University, Lahore — BS Artificial Intelligence 2024 – 2028

EXPERIENCE

AI Engineer Intern Jun 2026 – Jul 2026 | Remote
TEYZIX CORE

- Built a legal-tech contract-risk analyzer: a 7-stage async pipeline (FastAPI + Next.js 15) that extracts clauses, flags risk with confidence-scored explanations, answers questions via RAG over ChromaDB, and computes an explainable compliance score — shipped with role-based auth and an admin panel.
- Built TEYZIX, an AI document summarizer combining extractive scoring (frequency, TF-IDF, positional ranking) with abstractive BART (facebook/bart-large-cnn), adding multi-document batching and auto-detection across 16+ languages.

AI Developer Intern Mar 2026 – May 2026 | Paid | Remote
Punjab Skill Development Authority (PSDA)

- Sole engineer delivering a full-stack SaaS platform (Next.js + Tailwind + PostgreSQL) from Figma specs to production — public site, admin dashboard, and a 3-role permission system across 8+ REST API endpoints.
- Built an OCR document parser using the Gemini API to extract and structure text from multi-page uploads, integrated into the SaaS backend pipeline.

Software Engineer Intern Sep 2025 – Jan 2026 | Part-time | Remote
Artificial Automation Business Solutions (AABS)

- Built a RAG-based PDF chatbot with embedding-model selection, ChromaDB vector indexing, and retrieval-augmented generation, improving answer grounding over a naive keyword-search baseline.
- Developed a responsive SaaS dashboard UI in Next.js 15 / React 19 / TypeScript, including authentication, analytics visualizations, and subscription-management interfaces.

PROJECTS

Quality Evaluation PDF Chatbot — Production RAG pipeline with a hard RAGAS quality gate, hybrid search (dense + BM25 + reranking), and Prometheus observability. Python, FastAPI, ChromaDB, Groq/Llama 3.

Legal Document Risk Analyzer — Contract-risk analyzer with OCR ingestion, clause/risk extraction, RAG-grounded Q&A, and an explainable compliance score, behind role-based auth with an admin panel. FastAPI, Next.js 15, ChromaDB, Groq.

RecruitAI — Explainable resume-screening engine ranking candidates on a 100-point rubric (skills, experience, education, AI fit) with ChromaDB semantic skill matching. FastAPI, Next.js 15, Groq.

Agentic RAG PDF Chatbot — Full-stack PDF chatbot with agentic multi-query RAG, SSE token streaming, JWT per-user workspaces, and page-level citations. FastAPI, Next.js, Groq, pgvector.

Hadith Chatbot — 39,038 authentic Hadiths served via a two-stage BM25 → FAISS semantic re-ranking pipeline with LangChain orchestration for precise cross-Hadith attribution.

KRR Application — AI research platform for PDF ingestion, structured summarization, cross-paper analysis, and pgvector semantic search. FastAPI, Next.js, Groq, Docker.

TECHNICAL SKILLS

AI & Machine Learning	Python • PyTorch • Scikit-learn • OpenCV • RAG Systems • LangChain • LangGraph • ChromaDB • FAISS • pgvector • Sentence Transformers • Groq API • Gemini API • OpenAI API • LLM Evaluation (RAGAS) • Prompt Engineering
Web & Backend	Next.js • React • FastAPI • TypeScript • JavaScript • Tailwind CSS • PostgreSQL • SQLAlchemy • JWT Auth • Firebase • Streamlit • REST APIs
Tools	Git/GitHub • Linux • Docker • CI/CD • OCR (Tesseract) • Claude Code • Data Structures